1 There are some red counters and some blue counters in a bag.

The ratio of red counters to blue counters is 3:1.

Two counters are removed at random.

The probability that both the counters taken are blue is $\frac{1}{20}$

Work how many counters were in the bag before any counters were removed.

R	B	Total
$3x$	x	$4x$

$P(B, B) = 1/20$

$\Rightarrow x/4x \times \frac{(x-1)}{(4x-1)} = 1/20$

$\Rightarrow \frac{x-1}{4(4x-1)} = 1/20$

$\Rightarrow 20(x-1) = 4(4x-1)$

$\div 4 \qquad \div 4$

$\Rightarrow 5(x-1) = 4x-1$

$\Rightarrow 5x-5 = 4x-1$

$\Rightarrow 5x-4x = -1+5$

$x = 4$

Total $= 4x = 4 \times 4 = 16$

.....

(Total for question 1 is 5 marks)

2 There are some red counters and some blue counters in a bag.

The ratio of red counters to blue counters is 4:1.

Two counters are removed at random.

The probability that both the counters taken are red is $\frac{22}{35}$

Work how many blue counters are in the bag.

R	B	T
4x	x	5x

$P(R,R) = 22/35$

$\Rightarrow \frac{4x}{5x} \times \frac{(4x-1)}{(5x-1)} = 22/35$

$\Rightarrow \frac{16x^2 - 4x}{25x^2 - 5x} = \frac{22}{35} \Rightarrow \frac{4x(4x-1)}{5x(5x-1)} = 22/35$

$\Rightarrow 4(4x-1) \times 35 = 22 \times 5(5x-1)$

$\div 2$

$\Rightarrow 2x(4x-1) \times 35 = 11 \times 5 \times (5x-1)$

$\div 5$

$\Rightarrow 2 \times (4x-1) \times 7 = 11 \times 1 \times (5x-1)$

$\Rightarrow 14(4x-1) = 11(5x-1)$

$\Rightarrow 56x - 14 = 55x - 11$

$\Rightarrow 56x - 55x = -11 + 14$

$x = 3$

.....

(Total for question 2 is 5 marks)

3 There are 5 red counters and y blue counters in a bag.

Imogen takes a counter from the bag at random.
She puts the counter back into the bag.
Imogen then takes another counter at random from the bag.

The probability that the first counter Imogen takes is red and the second counter Imogen takes is red is $\frac{1}{9}$

Work how many blue counters are in the bag.

R / 5 b / y T / 5+y

P(r, r) = 1/9

$\Rightarrow \frac{5}{5+y} \times \frac{5}{5+y} = 1/9$

$\Rightarrow 25/y^2+25+10y \Rightarrow 25/y^2+25+10y = 1/9$

$\Rightarrow 225 = y^2+25+10y$

$-225 \qquad -225$

$\Rightarrow y^2 - 200 + 10y = 0$

$\Rightarrow y^2 + 20y - 10y - 200 = 0$

$\Rightarrow y(y+20) - 10(y+20) = 0$

$\Rightarrow (y-10)(y+20) = 0$

$y = 10 \qquad y = -20$

.....

(Total for question 3 is 5 marks)

4 There are 4 red counters and x blue counters in a bag.

2 counters are removed from the bag at random.

The probability that both the counters taken are blue is $\frac{1}{3}$

Work out the value of x.

R b T

4 x 4+x

$P(b,b) = 1/3$

$$\Rightarrow \frac{x}{4+x} \times \frac{(x-1)}{(\cancel{4+x-1})} = 1/3$$

$3\cancel{4}+x$

$$\Rightarrow \frac{x^2-x}{x^2+7x+12} = 1/3$$

$$\Rightarrow 3x^2-3x = x^2+7x+12$$

$$\Rightarrow 2x^2-10x-12 = 0$$

$$\Rightarrow x^2-5x-6 = 0$$

$$\Rightarrow x^2-3x-3x-6 = 0$$

$$\Rightarrow x(x+2)+3(x+2)$$

$$\Rightarrow (x-3)(x$$

n cunt

$24 \quad -6$

$-12 \quad -2+3$

$+6-1$

6

$-3+2$

$$\Rightarrow x^2+6x-1x-6$$

$$\Rightarrow x(x-1)+6(x-1)$$

$$\Rightarrow (x+6)(x-1) = 0$$

$$\Rightarrow x=-6 \quad x=1$$

(6)

(Total for question 4 is 6 marks)

5 There are 5 red counters and x blue counters in a bag.

2 counters are removed from the bag at random.

The probability that both the counters taken are red is $\frac{5}{33}$

Work out the value of x.

$\frac{R}{5}$ $\frac{B}{x}$ $\frac{T}{5+x}$

$P(R,R) = 5/33$

$\Rightarrow \frac{5}{5+x} \times \frac{4}{4+x} = \frac{20}{x^2+9x+20}$

$\Rightarrow \frac{20}{x^2+9x+20} = \frac{5}{33}$

$\Rightarrow \frac{660}{5x^2+45x+100} = \frac{660}{660}$

$\Rightarrow 5x^2+45x-550 = 0$

$\div 5$

$\Rightarrow x^2+9x-110 = 0$

-550

100
-660
-550

-110
$1 \overline{)110}$
$55\ 2$
$5\ 22$
$1\ \ 110$

.....

(Total for question 5 is 7 marks)

6 There are n counters in a bag.

4 of the counters are red and the rest are blue.

Ross takes a counter from the bag at random and does not replace it.
He then takes another counter at random from the bag.

The probability that Ross takes two blue counters is $\frac{1}{3}$

(a) Show that $n^2 - 13n + 30 = 0$

r	b	T
4	n-4	n

$P(b, b) = 1/3$

(b) Find the value of n.

(5)

.....

(2)

(Total for question 6 is 7 marks)

There are n counters in a bag.

8 of the counters are red and the rest are blue.

Adam takes a counter from the bag at random and does not replace it.
He then takes another counter at random from the bag.

The probability that Adam takes two blue counters is $\frac{1}{5}$

(a) Show that $n^2 - 21n + 90 = 0$

(5)

(b) Find the value of n.

.....

(2)

(Total for question 1 is 7 marks)